

A short description of the final project, and what it accomplished.

Our final project is a Marvel themed database application that allows users to manage and organize information about superheroes, villains, powers, teams, movies, universes and other Marvel related entities. It captures relationships between these entities, such as heroes having powers, characters appearing in movies, and having members in teams and organizations. The user interface is clear and allows users to perform basic database operations (insert, update, delete, search, project, join) and advanced operations (aggregation with group by, nested aggregation with group by, aggregation with having, and division).

How the final schema differed from the schema you turned in, and why:

The final schema is mostly unchanged from the schema submitted in Milestone 2. Some minor changes were made during implementation, such as renaming some relations for consistency with the final SQL script. For instance, in StoneColour, milestone 2 used Name, while the final SQL uses StoneName. Next, the Power attribute Type was changed to WeaponType. Along with this, some foreign key references and other attribute naming was cleaned up to match the implemented tables consistently. These changes were made to increase clarity around what the data represents.

A visual representation of the state of your database after running the .sql file.

AppearsIn Table:

	HEROACTORNAME	HEROALIAS	VILLAINACTORNAME	VILLAINALIAS	MOVIENAME	MOVIERELE
1	Chris Evans	CaptainAmerica	Hugo Weaving	RedSkull	Captain America: The First Avenger	2011
2	Chris Evans	CaptainAmerica	Tom Hiddleston	Loki	The Avengers	2012
3	Chris Hemsworth	Thor	Tom Hiddleston	Loki	The Avengers	2012
4	Chris Hemsworth	Thor	Tom Hiddleston	Loki	Thor	2011
5	Robert Downey Jr.	IronMan	Tom Hiddleston	Loki	The Avengers	2012

Superhero Table:

	ACTORNAME	ALIAS	CHARACTERNAME	STANDING	SPECIES	PUBLICIDENTITY
1	Robert Downey Jr.	IronMan	Tony Stark	Dead	Human	Known
2	Chris Evans	CaptainAmerica	Steve Rogers	Retired	Human	Known
3	Chris Hemsworth	Thor	Thor Odinson	Alive	Asgardian	Known
4	Mark Ruffalo	Hulk	Bruce Banner	Alive	Human	Known
5	Scarlett Johansson	BlackWidow	Natasha Romanoff	Dead	Human	Unknown
6	Tom Holland	SpiderMan	Peter Parker	Active	Human	Secret
7	Vin Diesel	Groot	Groot	Active	Flora Colossus	Unknown
8	Bradley Cooper	Rocket	Rocket Raccoon	Active	Genetically Modified Raccoon	Unknown
9	Paul Rudd	AntMan	Scott Lang	Active	Human	Secret
10	Elizabeth Olsen	ScarletWitch	Wanda Maximoff	Unknown	Enhanced Human	Known

Villain Table:

	ACTORNAME	ALIAS	CHARACTERNAME	STANDING	SPECIES	PUBLICIDENTITY	REASON
1	Tom Hiddleston	Loki	Loki Laufeyson	Alive	Frost Giant	Known	Power
2	Josh Brolin	Thanos	Thanos	Dead	Titan	Known	Overpopulation
3	James Spader	Ultron	Ultron	Destroyed	AI	Known	Evolution
4	Michael B. Jordan	Killmonger	Erik Stevens	Dead	Human	Known	Revenge
5	Cate Blanchett	Hela	Hela	Dead	Asgardian	Known	Conquest
6	Julian McMahon	DoctorDoom	Victor Von Doom	Alive	Human	Known	Power
7	Hugo Weaving	RedSkull	Johann Schmidt	Dead	Enhanced Human	Known	Domination
8	Willem Dafoe	GreenGoblin	Norman Osborn	Dead	Human	Known	Insanity
9	Michael Keaton	Vulture	Adrian Toomes	Alive	Human	Known	Profit

VillainHasPower Table:

	POWERID	VILLAINACTORNAME	VILLAINALIAS	DATEGAINED
1	11	Tom Hiddleston	Loki	04/05/11
2	12	Josh Brolin	Thanos	27/04/18
3	13	James Spader	Ultron	01/05/15
4	14	Michael B. Jordan	Killmonger	16/02/18
5	15	Cate Blanchett	Hela	03/11/17

WeaponSource Table:

	WEAPON	SOURCE
1	Mjolnir	Asgard
2	Shield	Vibranium
3	ArcReactor	Tech
4	Widow Bite	S.H.I.E.L.D.
5	Stormbreaker	Nidavellir
6	Necrosword	Nidavellir
7	WebShooter	Nidavellir
8	AntSuit	Nidavellir
9	Body	Nidavellir
10	ChaosEnergy	Nidavellir
11	MindStone	Nidavellir
12	Suit	Nidavellir
13	Marksmanship	Nidavellir
14	Bow	Nidavellir
15	StealthSuit	Nidavellir
16	Scepter	Nidavellir
17	InfinityGauntlet	Cosmic Strength
18	VibraniumBody	Synthetic
19	PantherHabit	Tech

Universe Table:

	EARTHNUMBER	NAME	STATUS	TIMELINE
1	616	Earth-616	Active	Main
2	199999	Earth-199999	Active	MCU
3	838	Earth-838	Destroyed	Alternate
4	2149	Earth-2149	Destroyed	Zombie
5	1610	Earth-1610	Destroyed	Ultimate

Avenger Table:

	ACTORNAME	ALIAS	AVAILABILITYLEVEL	DATEJOINED
1	Robert Downey Jr.	IronMan	5	2012
2	Chris Evans	CaptainAmerica	5	2012
3	Chris Hemsworth	Thor	5	2012
4	Mark Ruffalo	Hulk	4	2012
5	Scarlett Johansson	BlackWidow	4	2012

BasedIn Table:

	LOCATIONNAME	ORGANIZATIONNAME
1	Asgard	S.H.I.E.L.D.
2	Knowhere	Ten Rings
3	Sokovia	HYDRA
4	Wakanda	Wakandan Royal Guard

ExistsIn Table:

	TEAMNAME	EARTHNUMBER
1	Asgardians	838
2	Avengers	199999
3	Guardians of the Galaxy	616
4	Guardians of the Galaxy	199999
5	Young Avengers	616

Features Table:

	EARTHNUM	INFINITYSTONENAME	TEAMNAME	MOVIENAME	MOVIERELEASEYEAR
1	616	Mind Stone	Avengers	Avengers: Age of Ultron	2015
2	616	Power Stone	Guardians of the Galaxy	Guardians of the Galaxy	2014
3	616	Reality Stone	Guardians of the Galaxy	Guardians of the Galaxy	2014
4	616	Space Stone	Avengers	The Avengers	2012
5	616	Time Stone	Masters of the Mystic Arts	Doctor Strange	2016

Fights Table:

	HEROACTORNAME	HEROALIAS	VILLAINACTORNAME	VILLAINALIAS
1	Chris Evans	CaptainAmerica	Tom Hiddleston	Loki
2	Chris Hemsworth	Thor	Cate Blanchett	Hela
3	Mark Ruffalo	Hulk	James Spader	Ultron
4	Robert Downey Jr.	IronMan	Josh Brolin	Thanos
5	Scarlett Johansson	BlackWidow	Michael B. Jordan	Killmonger

HeroHasPower Table:

	HEROACTORNAME	HEROALIAS	POWERID	DATEGAINED
1	Chris Evans	CaptainAmerica	1	22/07/11
2	Tom Holland	SpiderMan	2	07/07/16
3	Paul Rudd	AntMan	3	17/07/15
4	Vin Diesel	Groot	4	01/08/14
5	Elizabeth Olsen	ScarletWitch	5	01/05/15

InfinityStone Table:

	EARTHNUMBER	NAME	STATUS	OWNER
1	616	Space Stone	Destroyed	Thanos
2	616	Mind Stone	Destroyed	Vision
3	616	Reality Stone	Destroyed	Collector
4	616	Power Stone	Destroyed	Nova Corps
5	616	Time Stone	Destroyed	Doctor Strange

LocatedIn Table:

	TEAMNAME	LOCATIONNAME
1	Asgardians	Sokovia
2	Avengers	Wakanda
3	Guardians of the Galaxy	Knowhere
4	Young Avengers	Sokovia
5	Young Avengers	Wakanda

Location Table:

	NAME	SPECIES	PLANET	POPULATION
1	Titan	Titanian	Titan	Extinct
2	Xandar	Xandarian	Xandar	Unknown
3	Ego	Celestial	Cosmic	1
4	Vormir	Unknown	Vormir	Unknown
5	New York	Human	Earth	8400000
6	Kamar-Taj	Human	Earth	Unknown
7	Wakanda	Human	Earth	6000000
8	Asgard	Asgardian	Asgard	Unknown
9	Knowhere	Various	Cosmic	Unknown
10	Sokovia	Human	Earth	Unknown

MemberOfOrganization Table:

	HEROACTORNAME	HEROALIAS	VILLAINACTORNAME	VILLAINALIAS	ORGANIZATIONNAME
1	Chris Evans	CaptainAmerica	Tom Hiddleston	Loki	S.H.I.E.L.D.
2	Chris Hemsworth	Thor	Cate Blanchett	Hela	Wakandan Royal Guard
3	Mark Ruffalo	Hulk	James Spader	Ultron	Stark Industries
4	Robert Downey Jr.	IronMan	Josh Brolin	Thanos	Stark Industries
5	Scarlett Johansson	BlackWidow	Michael B. Jordan	Killmonger	S.H.I.E.L.D.

MemberOfTeam Table:

	ACTORNAME	ALIAS	NAME
1	Chris Evans	CaptainAmerica	Avengers
2	Chris Hemsworth	Thor	Asgardians
3	Elizabeth Olsen	ScarletWitch	Avengers
4	Paul Rudd	AntMan	Avengers
5	Tom Holland	SpiderMan	Avengers

Movie Table:

	NAME	RELEASEYEAR	NUMBEROFORDER	ROTTENTOMATO
1	Iron Man	2008	1	94
2	The Incredible Hulk	2008	2	67
3	Thor	2011	3	77
4	Captain America: The First Avenger	2011	4	80
5	The Avengers	2012	5	91
6	Avengers: Age of Ultron	2015	8	87
7	Guardians of the Galaxy	2014	6	91
8	Doctor Strange	2016	7	91

MovieOrderPhase Table:

	NUMBEROFORDER	PHASE
1	1	1
2	2	1
3	3	1
4	4	1
5	5	2
6	6	2
7	7	1
8	8	1

Organization Table:

	NAME	LEADER	STATUS	TYPE	FOUNDER
1	S.H.I.E.L.D.	Nick Fury	Active	Intelligence	Howard Stark
2	HYDRA	Red Skull	Inactive	Terrorist	Johann Schmidt
3	Stark Industries	Tony Stark	Active	Corporation	Howard Stark
4	Wakandan Royal Guard	Okoye	Active	Military	T'Chaka
5	Ten Rings	Xu Wenwu	Active	Criminal	Xu Wenwu

Power Table:

	ID	SKILLSET	WEAPON	WEAPONTYPE
1	1	Super Strength	Shield	Vibranium
2	2	Web Slinging	WebShooter	Tech
3	3	Size Manipulation	AntSuit	Tech
4	4	Regeneration	Body	Biological
5	5	Magic	ChaosEnergy	Mystic
6	6	Telekinesis	MindStone	InfinityStone
7	7	Flight	Suit	Tech
8	8	Marksmanship	Bow	Weapon
9	9	Invisibility	StealthSuit	Tech
10	10	Energy Blasts	ArcReactor	Tech
11	11	Illusion Magic	Scepter	Mystic
12	12	Cosmic Strength	InfinityGauntlet	Cosmic
13	13	Artificial Intelligence	VibraniumBody	Synthetic
14	14	Enhanced Combat	PantherHabit	Tech
15	15	Death Magic	Necrosword	Mystic

SetIn Table:

	EARTHNUMBER	MOVIENAME	MOVIERELEASEYEAR
1	199999	Captain America: The First Avenger	2011
2	199999	Iron Man	2008
3	199999	The Avengers	2012
4	199999	The Incredible Hulk	2008
5	199999	Thor	2011

StoneColour Table:

	STONENAME	COLOUR
1	Space Stone	Blue
2	Mind Stone	Yellow
3	Reality Stone	Red
4	Power Stone	Purple
5	Time Stone	Green

TakesPlace Table:

	MOVIENAME	MOVIERELEASEYEAR	LOCATIONNAME
1	Captain America: The First Avenger	2011	Wakanda
2	Iron Man	2008	Wakanda
3	The Avengers	2012	Knowhere
4	The Incredible Hulk	2008	Sokovia
5	Thor	2011	Asgard

Team Table:

	NAME	NUMBEROFPEOPLE	STANDING	MENTOR	LEADER
1	Avengers	6	Active	Nick Fury	Captain America
2	Guardians of the Galaxy	5	Active	Yondu	Star-Lord
3	Young Avengers	5	Active	Hawkeye	Kate Bishop
4	Asgardians	5	Active	Fandral	Fandral
5	Masters of the Mystic Arts	5	Active	Doctor Strange	Doctor Strange
6	Defenders	4	Inactive	Stick	Daredevil
7	Fantastic Four	4	Active	Reed Richards	Mr. Fantastic

List of non-hardcoded and hardcoded SQL queries:

Non-hardcoded Queries 1-6:

Insert Query

SQL:

```
INSERT INTO HeroHasPower (HeroActorName, HeroAlias, PowerID, DateGained)
VALUES (:heroActorName, :heroAlias, :powerID, TO_DATE(:dateGained, 'YYYY-MM-DD'));
```

Location: 183-186 appService.js

Update Query

SQL:

```
UPDATE Power
SET ${updates.join(", ")}
WHERE ID = :powerID;
```

Location: 215-219 appService.js

Delete Query

SQL:

```
DELETE FROM Location
WHERE Name = :locationName;
```

Location: 309-312 appService.js

Selection Query

SQL:

```
SELECT *
FROM Superhero
WHERE <user-defined conditions>
```

Example:

```
SELECT *
FROM Superhero
WHERE Species = :value0 AND Standing != :value1
```

Location: 340-343 appService.js

Projection Query

SQL:

```
SELECT <user-selected attributes>
FROM Universe;
```

Example:

```
SELECT Name, Status, EarthNumber
FROM Universe
```

Location: 365-366 appService.js

Join Query

SQL:

```
SELECT s.ActorName,
       s.Alias,
       s.CharacterName,
       h.PowerID
FROM Superhero s
JOIN HeroHasPower h
  ON s.ActorName = h.HeroActorName
 AND s.Alias = h.HeroAlias
WHERE s.Species = :species
ORDER BY s.Alias, h.PowerID;
```

Location: 443-453 appService.js

Hardcoded Queries 7-10:

Aggregation with GROUP BY Query

SQL:

```
SELECT Standing,
       COUNT(*) AS Count
FROM Villain
GROUP BY Standing;
```

Location: 380-382 appService.js

Description: This counts how many villains fall under each standing.

Aggregation with HAVING

SQL:


```
SELECT Species,  
        COUNT(*) AS numHeroes  
FROM Superhero  
GROUP BY Species  
HAVING COUNT(*) > 1  
ORDER BY numHeroes DESC;
```

Location: 395-401 appService.js

Description: This query groups superheroes by Species and counts how many heroes are in each group. It then filters the groups to only include species that have more than one superhero.

Nested aggregation with GROUP BY Query

SQL:

```
SELECT s.Species,  
        COUNT(*) AS HeroCount  
FROM Superhero s  
GROUP BY s.Species  
HAVING COUNT(*) >= ALL (  
        SELECT COUNT(*)  
        FROM Superhero s2  
        GROUP BY s2.Species  
);
```

Location: 470-479 appService.js

Description: This finds the species with the highest number of superheroes by comparing each species count to all the others.

Division Query

SQL:

```
SELECT s.ActorName,  
        s.Alias  
FROM Superhero s  
WHERE NOT EXISTS (  
        SELECT DISTINCT p.PowerID  
        FROM PowerFrom p
```

```
WHERE p.StoneName = 'Space Stone'
AND NOT EXISTS (
  SELECT 1
  FROM HeroHasPower h
  WHERE h.HeroActorName = s.ActorName
        AND h.HeroAlias = s.Alias
        AND h.PowerID = p.PowerID
)
);
```

Location: 415-429 appService.js

Description: Finds superheroes who possess all powers associated with the Space Stone.